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Re-executing Scientific Design after Method Changes

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Abstract

Changing a device, changing the method that designs it, and changing the procedure that improves that method invalidate different computations. We study these three interventions separately in an executable battery-free design apparatus. This development draft reports six deterministic invalidation controls and four fresh prompt-intervention attempts. Three delivered qualified computational designs; the fourth stopped at a frozen resource gate before validation. All 68 model requests settled. The method update remains unpromoted. A separate, explicitly continued battery-free episode produced three device programs. A queue-rank change passes fresh-reception gates but violates the frozen oldest-first requirement; the twelve-hour dark service remains infeasible in the fixed ideal design space. The product is not accepted. No method advantage, recursive self-improvement, unseen-task transfer, hardware validation, or speedup is established.

1 Question and scope

A previously successful device trace can cease to describe the system after a legal change. Reusing that trace may silently preserve obsolete control decisions. At the method level, a changed global design prompt may alter the first observation, the candidate, and every later experiment. Evaluating the old device under that prompt does not evaluate the changed method. Our question is which results remain valid after a typed intervention, and how to confirm a proposed method change without charging a cached response as a new design attempt.

2 Objects and validity contract

Let D contain device structure, circuits, firmware, storage, communication, and deployment. Let H contain the design prompt, skills, available observations, context policy, experiment order, and organization. Let G propose and select changes to H. Device runtime, the agent's decision span, and method generations are separate quantities. A record is reusable only when all consumed inputs, environment identities, numerical settings, and output hashes remain applicable. An evaluation change may reuse raw physical output while requiring a new verdict.

For a global prompt change $H \to H'$, our conservative rule restarts at task entry with a new owner, initial candidate, and context. Exact lower-level input responses may still be reused. A local continuation requires a separate proof that its restored state is valid. A G comparison additionally fixes the starting H_0, development distribution, selection protocol, and improvement budget.

3 F1a: device-response controls

We compare old fixed-trajectory replay, conservative dependency invalidation, boundary invalidation, full response execution, and exact-input response caching. The first qualification apparatus has energy arrivals, finite storage, a threshold controller, and load consumption. The independent full response is the oracle. In a feedback case, both dependency and boundary invalidation expand to the full response; this null advantage is a required control, not a speedup result.

In the executed deterministic controls, fixed-trajectory replay disagrees with the full response on 5 of 6 inputs. The conservative dependency, boundary, and exact-cache results agree on all 6. Only the unchanged input is an initial exact-cache hit. These are synthetic correctness controls; execution counts do not measure wall-time speedup.

Input	Replay packets	Full packets	Replay valid
unchanged	4	4	True
less-energy	4	2	False
higher-load	4	3	False
threshold-only	4	3	True
capacity-limit	4	0	False
arrival-order	4	2	False

4 F1b and early F4: real method re-execution

The frozen public task uses the already qualified piezoelectric, vendor PMIC, host-C controller, and digital receiver apparatus. The power board is fixed; the agent must revise firmware and deliver six valid frames in twenty seconds. The declared initial charge and current envelope remain assumptions. The task is an exposed development derivative of historical T1, not a fresh holdout.

H_0 requests one complete design attempt with execution, validation, decision, report, release, and close. H_1 adds a constraint and time-origin audit before a single justified revision, with version-bound evidence. Both receive the same full task and sources, backend, tool definitions, limits, and initial native cache. The prompt update was proposed by the research operator from earlier timing and tool-use failures; it is not an autonomous improver. Two attempts per condition run in order H_0, H_1, H_1, H_0. Each allows at most 80 parent-plus-child requests, four candidate evaluations, and 1,200 seconds of decision wall time. Organization is single-agent. No automatic resampling is allowed.

The protocol requires full execution to confirm an attempted method. Development retention requires both H_1 confirmations to deliver admissible results without an increased failure count. With two attempts per condition, cost and quality differences are descriptive. A tie, failure, or no retained update is a valid result.

Observed attempts and resource stop. All 4 fresh controllers consumed their assigned goal, started with zero prior trajectory messages, and changed the initial firmware. After administrative identifiers are normalized, the base prompts match; the harness, task, initial device, and physical cache start are identical. 3 attempts completed all six computational qualification gates and delivery. The last attempt, H_0-r2, completed its native numerical execution but hit the frozen 5 GB free-space gate before validation and release. It stopped after 260.958 seconds and was not resampled. Its pending qualification is not converted into a device failure or a pass.

Attempt	Requests	USD upper	Executions	Native	Frames	Outcome
H0-r1	13	0.188330	1	2	14	PASS
H1-r1	20	0.358763	2	2	10	PASS
H1-r2	20	0.361660	2	2	7	PASS
H0-r2	15	0.234168	2	2	–	STOP

Across all attempted owners, including the stop, 68 requests settled with no missing provider usage or unresolved reservation. Their conservative charge upper bound is USD 1.142922; the rate-based cache-aware estimate is USD 0.350731, not an invoice. The 7 candidate executions used 8 new native dispatches. Each newly solved input is disjoint from earlier successful native inputs; applicable exact inputs were reused. The sixteen-entry initial cache was copied without new solves. Its historical construction bill is unknown, not zero. Operator proposal and Codex supervision, executor-chat, and writing costs are outside the BF ledger; these attempt charges cannot establish end-to-end improvement efficiency.

Both H_1 confirmations delivered, using twenty requests each; the completed H_0-r1 used thirteen. The interrupted second H_0 prevents a completed arm comparison. We preserve H_1 as an experimental artifact, keep the default method unchanged, and make no success-rate, cost, speedup, transfer, or recursive improvement claim. Provider and OS caches were uncontrolled; setup and subsequent wall times remain separate in the audit.

5 A/R: a battery-free entry to device and method changes

The A/R case places an indoor-light-powered temperature and history tag on a transport container. Its public apparatus couples assumed photovoltaic port power, one capacitor, regulator thresholds, sensing and persistent writes, program activity, and a digital receiving endpoint. Source power is specified in watts; illuminance is metadata rather than a conversion to harvested power. All runs start with zero stored energy. The source, ideal storage and write models, and storage-side energy costs are uncalibrated computational assumptions.

5.1 The retained reference and legal intervention

We re-read the sealed reference without rerunning an unchanged design. All 81 sealed files and 17 original positive and negative conditions passed their original independent consumer. A phase-one consumer retains those outcomes and adds explicit gate applicability and maximum inter-sample spacing. The six-hour profile has 348 samples and its first fresh reception at 773 s. In the twelve-hour dark history profile, storage retains all 170 records, including 168 temperature samples across the whole profile. Yet 648 of the 720 historical dark-window bins contain no new sample. The recorded gap is delivered, but absolute time after reset remains unknown; all 73 queued records miss the recovery deadline and one of two required temperature events is missed.

Profile	Samples	New sampling	Retention	Recovery	Fresh delay
Six hours	348	Pass	Pass	N/A	Pass
Twelve-hour dark	168	Fail	Pass	Fail	Fail
Gateway recovery	168	Pass	Pass	Pass	Fail

Table 1: Independent readback of the original A/R device, before the Pi revisions. Recovery allows 120 s; fresh reception allows 10 s for each new online sample. N/A denotes an unexercised gate. Samples count the whole profile; hardware qualification is unknown for every row.

The original gateway-recovery profile separates backlog completion from timeliness: its 60 queued records are acknowledged within 82 s, and two repeated arrivals cause no duplicate acceptance, but one new sample waits 30 s. Preserved history, eventual upload, new sensing, and timely reception need separate evidence.

The frozen phase-one device space permits capacitance and the existing threshold domain, with maximum voltage 3.3 V and all source and load assumptions fixed. The participant can change sample, ACK and retry timing and supply a bounded numeric program that ranks pending records. This program observes sequence, record kind, and age only when synchronized local time and the record timestamp support it. It cannot observe true creation time after clock loss, future supply, or evaluator labels. Its expression is executed by the queue consumer; submission alone cannot create an acceptance receipt. The frozen task also requires the oldest unacknowledged record first. This ordering requirement must be checked separately from the rank interface's syntactic legality.

5.2 One actual episode with an explicit continuation

S0 dispatched Pi under owner `pi-chat-cf4b6cb2ebe9b771`. The provider reported `deepseek-flash`; this alias does not independently pin weights or a model version. The initial segment stopped before a device submission when SSE framing exceeded the host's 2 MB response envelope. It had made two provider requests and four inspection calls. The initially unresolved second request later settled. Its failure and original UNKNOWN accounting snapshot remain saved.

The explicit `continuation-001` resumed the same session, preserving its eight original messages and four tool results. It retained the original deadline and cumulative caps of 60 requests, three candidates and 3,600 s; the response envelope alone increased to 64 MiB. This is one continued episode, not an independent resample or a fresh method condition. The saved session prefix, model tool arguments, candidate JSON, source manifests, seals and settled provider rows are bound in the accompanying paper data. Hash readback distinguishes canonical JSON content hashes from file byte hashes, and the independent energy proof script from the service checker. Both checker identities match their respective receipts; all 68 original and 77 continuation bound files match.

Pi checked three adaptive device revisions, each on the same four public profiles: twelve new local case simulations and zero native dispatches. All use $C = 0.015$ F, off/ready thresholds 2.3/2.7 V, a 60 s sample period, 1 s ACK spacing, and rank $= 0 - \text{seq}$. The lowest score is selected, so this expression sends the newest pending record first. Retry intervals are 2, 30 and 8 s, respectively. Pi submitted candidate 002, preserving both rejected alternatives (Table 2). Its choice changes the executed queue program relative to the original device; this is a D intervention.

Candidate	Retry (s)	History samples	Late backlog	Missed events
001	2	106	10	2
002*	30	168	73	1
003	8	129	34	1

Table 2: Three adaptive device revisions in one continued Pi episode; * marks the submitted candidate. Samples count the entire history profile, including its light periods, rather than only the dark interval. Every listed backlog record misses the 120 s recovery deadline. The four history service failures persist for every candidate.

5.3 Conditional rank repair and the unresolved FIFO requirement

The independent consumer passes energy, actual activity/program replay, CRC and receiver checks for all twelve case executions. Under the rank interface, all three candidates pass applicable computational service gates in the cold, six-hour light/dark, and gateway recovery/deduplication profiles. In the gateway case they retain 168 samples, acknowledge the 60-record backlog within 120 s, and accept no duplicate from two repeated arrivals. Each of the 90 eligible new online samples is received without an extra simulation-step delay. This changes the fresh-delay gate from the original FAIL to PASS under the rank policy.

This conditional timing repair is not compliant with the frozen FIFO requirement. Independent pending-record replay finds strict ordering violations for all three candidates in both the history and gateway profiles. For the submitted candidate, the gateway profile's first witness occurs at 833 s: record 2 is selected while record 1 is still the oldest unacknowledged record. The rank hook permits that choice, but the original task text does not. The contract is unchanged and product acceptance remains `NOT_ACCEPTED`. A human must resolve this semantic conflict before any revised ordering contract and its results can be accepted.

All three history profiles still fail continuous new sampling, dark new sampling, 120 s recovery upload and temperature-event coverage; absolute time after reset remains UNKNOWN. Their whole-profile sample counts are 106, 168 and 129, not counts of samples taken entirely within darkness. Faster retries reduce the remaining late backlog while consuming energy that could support sensing. The selected candidate retains 170 records, including 168 samples, and reports the gap, but all 73 recovery backlog records are late and one of two events

is missed. Its fresh-delay gate passes only for the 78 eligible samples actually created during online service; it does not fill the missing dark observations.

5.4 A necessary bound and the D/H/G claim boundary

Pi had access to Codex-provided necessary inequalities, including a stronger sleep-plus-sensing dark bound. The final external proof independently uses the weaker sleep-only bound below, without calling that analytical tool. Granting all optimistic startup energy and omitting boot, sensing and transmission costs, startup by 1,800 s at the legal ready-voltage infimum of 2.2 V requires

$$C \le \frac{2(90 \cdot 0.8 - 1)\,10^{-6} \cdot 1800}{2.2^2} = 0.05281 \text{ F}.$$

In the fixed legal apparatus, the program cannot request power-off. Its fixed ESR cannot cause a transmission dropout above the smallest off threshold, and an ordinary brownout cannot recharge to a higher ready threshold in zero light. Continuous dark sampling therefore needs power between its first and last required samples. Grant full charge at dark entry and 122 s total edge slack (one 60 s interval plus a 1 s step at each end). Even charging only 5 μW sleep power requires

$$C \ge \frac{(43200 - 122) \cdot 5\,10^{-6}}{(3.3^2 - 2.2^2)/2} = 0.07120 \text{ F} > 0.05281 \text{ F}.$$

These disjoint necessary bounds establish a no-go only in the frozen ideal single-capacitor, fixed-sleep legal space. They are not physical impossibility or a participant discovery. Optimistic necessary limits suggest a cold deadline of at least 2,426.93 s (40.45 min), a dark duration of at most 32,072 s (8.91 h), or sleep power below about 3.7084 μW. None is a sufficient design. Any changed service or apparatus needs a new human-approved contract and qualification.

The episode totals seven SETTLED provider requests, including the two from the stopped segment. Their charge upper bound is USD 0.082605 and the cache-aware rate estimate is USD 0.058482888; neither is an invoice. No provider usage is unresolved at final readback. Codex apparatus, checker and repair assistance remains `UNPRICED_NONZERO`, so the full end-to-end cost is unknown. The earlier four H attempts, their 68 settled requests, all failures and all historical UNKNOWN records remain separate and unchanged.

The actual episode supplies a device-level conditional rank repair and a negative service result. A proposed H rule could check joint startup/dark feasibility and distinguish backlog completion from fresh delay, but no fresh H_0/H_1 comparison or retained update is reported for A/R. A procedure that proposes and selects that rule would change G and need controlled lineages and future tasks. Method benefit, recursive self-improvement and held-out transfer are untested here; hardware qualification remains UNKNOWN.

6 Costs, exposure, and competing explanations

Report all parent and child requests, settled usage, reservations, failed tools, native dispatches, retries, confirmations, cache construction, and unresolved UNKNOWN separately. Total cost is

$$C = C_{\text{proposal}} + C_{\text{selection}} + C_{\text{confirmation}} + C_{\text{failures}} + C_{\text{cache construction}}.$$

A valid cache hit does not remove the cost of building its entry. Paired attempts start from identical cache manifests rather than warming later conditions with earlier outputs. Provider identity and usage availability are dispatch gates. Preflight failures remain recorded, and cannot become successful attempts by silent replacement. Mechanism exposure is shared with the interface study.

F1a supports a correctness boundary for this small control; it does not establish a general incremental algorithm. F1b can establish that a prompt was actually changed and new design decisions executed. Favorable development outcomes would still admit task familiarity, stochastic variation, and operator selection as competing explanations. Later selection and final tasks must be frozen separately.

7 Related work and remaining experiments

Executable build-system decompositions provide a strong dependency-invalidation baseline [1]. Prompt optimizers such as GEPA propose and evaluate prompt changes using execution feedback [2]. Our current work does not establish superiority over either family. The planned study isolates physical-response reuse from full method confirmation and retains negative cases. F2/F3 will explain decision spans and organizations on a useful subset. F5 requires independent G_0/G_1 lineages from a fixed H_0; no such lineage experiment is reported. F6 will follow the continuous A/R product after its service reference is qualified. Physical calibration and hardware acceptance remain separate work.

References

[1] Andrey Mokhov, Neil Mitchell, and Simon Peyton Jones. Build systems a la carte. ICFP, 2018. https://www.microsoft.com/en-us/research/publication/build-systems-la-carte/.

[2] Lakshya A. Agrawal et al. GEPA: Reflective Prompt Evolution Can Outperform Reinforcement Learning. arXiv:2507.19457v2, 2026. https://arxiv.org/abs/2507.19457v2.